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Regla de la cadena para derivadas de Wirtinger

Proposición 1 (Regla de la cadena). *Sean* $f \in \mathcal{C}^1(\Omega_1)$ *y* $g \in \mathcal{C}^1(\Omega_2)$ *con* $f(\Omega_1) \subset \Omega_2$, *entonces se satisfacen*

(1) $\dfrac{\partial}{\partial z}(g \circ f)(z) = \dfrac{\partial g}{\partial z}\,(f(z))\,\dfrac{\partial f}{\partial z}(z) + \dfrac{\partial g}{\partial \overline{z}}\,(f(z))\,\dfrac{\partial \overline{f}}{\partial z}(z).$

(2) $\dfrac{\partial}{\partial \overline{z}}(g \circ f)(z) = \dfrac{\partial g}{\partial z}\,(f(z))\,\dfrac{\partial f}{\partial \overline{z}}(z) + \dfrac{\partial g}{\partial \overline{z}}\,(f(z))\,\dfrac{\partial \overline{f}}{\partial \overline{z}}(z).$

***Demostración*:** Veremos únicamente la primera identidad, la segunda se demuestra de forma análoga. Escribimos $f(z) = u(x,y) + i\,v(x,y)$. Entonces, por definición de derivada de Wirtinger y la regla de la cadena para derivadas parciales, se tiene que

$$2\,\frac{\partial}{\partial z}(g \circ f)(z) = \left(\frac{\partial}{\partial x} - i\,\frac{\partial}{\partial y}\right)(g \circ f)(z) = \frac{\partial g}{\partial u}\,\frac{\partial u}{\partial x} + \frac{\partial g}{\partial v}\,\frac{\partial v}{\partial x} - i\frac{\partial g}{\partial u}\,\frac{\partial u}{\partial y} - i\frac{\partial g}{\partial v}\,\frac{\partial v}{\partial y} = (*).$$

Aplicando ahora el `lem-derivadas-parciales-wirtinger`/Lema 1, tenemos que

$$\begin{aligned}
(*) &= \left(\frac{\partial g}{\partial z} + \frac{\partial g}{\partial \overline{z}}\right)\frac{\partial u}{\partial x} + i\left(\frac{\partial g}{\partial z} - \frac{\partial g}{\partial \overline{z}}\right)\frac{\partial v}{\partial x} - i\left(\frac{\partial g}{\partial z} + \frac{\partial g}{\partial \overline{z}}\right)\frac{\partial u}{\partial y} + \left(\frac{\partial g}{\partial z} - \frac{\partial g}{\partial \overline{z}}\right)\frac{\partial v}{\partial y} \\
&= \frac{\partial g}{\partial z}\left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} - i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}\right) + \frac{\partial g}{\partial \overline{z}}\left(\frac{\partial u}{\partial x} - i\frac{\partial v}{\partial x} - i\frac{\partial u}{\partial y} - \frac{\partial v}{\partial y}\right) \\
&= \frac{\partial g}{\partial z}\left(2\frac{\partial u}{\partial z} + 2i\frac{\partial v}{\partial z}\right) + \frac{\partial g}{\partial \overline{z}}\left(2\frac{\partial u}{\partial \overline{z}} + 2i\frac{\partial v}{\partial \overline{z}}\right) = 2\,\frac{\partial g}{\partial z}(f(z))\,\frac{\partial f}{\partial z}(z) + 2\,\frac{\partial g}{\partial \overline{z}}(f(z))\,\frac{\partial \overline{f}}{\partial z}(z).
\end{aligned}$$

Luego el resultado se sigue dividiendo entre 2. ■

Referenciado en

- `lem-laplaciano-wirtinger`
- `prop-fn-holomorfa-iff-wirtinger`
- `lem-derivadas-parciales-wirtinger`
- `cor-wirtinger-composicion-holomorfas`

- `prop-regla-cadena-wirtinger`